

Single-neuron response strength and latency predict held-out group currents

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Abstract

Public cortical-slice recordings pair two-photon stimulation of individual neurons with stimulation of groups of neurons while postsynaptic current is recorded from one patched cell. These data were used to test whether single-neuron responses predict held-out group responses. For each laser power, a response shared across neurons was estimated first. Repeatable neuron-specific differences were then learned from two halves of the single-neuron trials and added to predict group trials; no group response was used for fitting.

The rule was developed in nine somatostatin-to-excitatory (SST-to-E) files and frozen before the excitatory-to-excitatory (E-to-E) files were opened. The E-to-E cohort contained 129,613 single-neuron trials and 74,435 group trials across ten recorded cells. The reliability-weighted model improved demixed-current mean-squared error (MSE) over a power-only baseline in all ten files (mean gain 2.12×10^{-5} , $q = 0.00195$). Correct neuron identity was necessary in shuffle and replacement controls. A post-feedback comparison showed that source-specific response strength explained 96.7% of the MSE improvement; adding a source-specific latency shift explained nearly all of it. The full waveform improved correlation over power alone, but not conclusively over strength plus latency. In raw current, the full waveform did not improve correlation, but the simpler gain-plus-latency model did so in all ten files in the post-feedback analysis. The public files do not identify animal or slice membership, so inferential tests are reported at the file level and a three-date clustering is included as a conservative sensitivity analysis.

Keywords: two-photon holographic optogenetics; postsynaptic current; single-neuron stimulation; held-out prediction; source-specific response strength; response latency; neural waveform demixing

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1 Introduction

Observing neural activity does not by itself show what a neuron contributes when it is activated. Two-photon holographic optogenetics provides a direct test: selected neurons can be stimulated individually or together while the current entering a patched postsynaptic neuron is recorded [Packer et al., 2012, 2015, Mardinly et al., 2018, Adesnik and Abdeladim, 2021].

The public data analyzed here come from cortical slices [Gajowa and Triplett, 2025, Triplett et al., 2025]. Each file represents one patched postsynaptic cell and contains hundreds of candidate neurons distributed across four or five imaging depths. On each trial, a matrix records which candidates were stimulated and at what laser power. The file also contains the resulting 45-ms current trace in raw form and after neural waveform demixing (NWD). Trials stimulating one candidate can therefore be used for learning, while trials stimulating groups can be reserved for evaluation.

The source study used CAViaR to predict a held-out hologram after fitting responses to other group holograms [Triplett et al., 2025]. The contrast here is narrower: no group response is used for fitting. The question is whether responses learned only from single-neuron trials transfer when those neurons are stimulated in groups.

Let s identify the stimulated neuron, p the laser power, and t the time after stimulation. Repeated trials showed that some apparent differences between neurons were unstable. The model keeps only the repeatable part of the single-neuron response:

$$r_{s,p}(t) = g_p(t) + \lambda_p d_{s,p}(t), \quad (1)$$

$$\hat{y}_E(t) = \sum_{s \in E} r_{s,p_s}(t), \quad (2)$$

Here, $g_p(t)$ is the response shared across neurons at power p , $d_{s,p}(t)$ is the difference for neuron s , and λ_p measures how well those differences repeat across two halves of the single-neuron trials (Fig. 1). Unreliable differences are reduced and reliable differences are retained. The model then adds the learned responses of all stimulated neurons. Group responses are used only for testing, never for fit-

Table 1: Terminology. Definitions used throughout the paper.

Term	Meaning in this study
Single-neuron trial	A trial in which exactly one neuron is stimulated; these trials are used to learn responses.
Group trial	A trial in which several neurons are stimulated together; these trials are held out to test predictions.
Shared response	The part of the waveform that is common across neurons stimulated at the same laser power.
Neuron-specific difference	The part that distinguishes one stimulated neuron from the shared response.
Reliability λ_p	The fraction of neuron-specific detail kept because it repeats across two halves of the single-neuron trials.
Recording file	One patched postsynaptic cell, its candidate stimulated neurons, the stimulation design, and the recorded current trials.
Demixed current	Current after neural waveform demixing (NWD), a source-identity-blind U-Net that suppresses overlapping events and noise.
Power-only baseline	A prediction that knows the laser power of each stimulus but not which neurons were stimulated.

ting.

The order of the analysis matters. A planned first model improved MSE but missed its waveform-correlation criterion in SST-to-E recordings. The already-open SST data were then used to develop the reliability rule. The new rule was frozen before the E-to-E dataset was downloaded or inspected. The first dataset was used for development; the second provided a prospectively held-out test at the recording-file level.

2 Results

2.1 The recordings pair learning trials with held-out group trials

The nine SST-to-E files record stimulation of somatostatin neurons while current is measured from an excitatory postsynaptic cell. The ten E-to-E files record stimulation of excitatory neurons while current is measured from another excitatory cell. Each file contains one recorded postsynaptic cell, 214–687 candidate stimulated neurons, four or five imaging depths, a trial-by-neuron stimulation matrix, and raw and NWD-demixed current. Across the E-to-E cohort there were 4,724 candidate-neuron entries and 204,054 trials. Exactly one neuron was stimulated in 129,613 trials; more than one was stimulated in 74,435 trials; six blank trials were excluded.

The distinction between trial types defines the prediction task. Single-neuron trials estimate each neuron’s response. Group trials test the sum of those estimates. A group trial is eligible only when every stimulated neuron has at least three single-neuron trials at the exact delivered power. All 74,435 E-to-E group trials met this rule (Table 2).

2.2 The first test revealed unstable waveform detail

The first test used nine checksum-verified SST-to-E files with 46,578 trials. Of these, 26,803 stimulated one neuron and 19,769 stimulated groups. A total of 19,217 group trials had enough single-neuron data to make a prediction. The first model added the full mean waveform for each stimulated neuron. The baseline ignored neuron identity and added the shared mean response at each delivered power.

Using neuron identity improved demixed response-window MSE over the baseline in all nine files (mean file gain 8.78×10^{-4} , exact $p = 0.001953$, Benjamini–Hochberg $q = 0.003906$). However, it did not improve waveform correlation (mean $\Delta\rho = -0.00263$, $p = 0.682$, $q = 0.682$). The plan required both measures to improve, so fourteen PV-to-E files remained unopened (Fig. 2). Neuron identity still carried information about response size. The failed waveform criterion motivated separating repeatable neuron-specific detail from trial-to-trial noise.

2.3 Repeatability identified which details to keep

An exploratory follow-up tested whether repeatability identified neuron-specific responses that improved group-response prediction. Median split-half reliability closely tracked the gain in waveform correlation across the nine files (Spearman $\rho = 0.957$). This observation was exploratory because the SST group responses had already been opened. For the new rule, alternating trials provided two independent estimates of the neuron-specific differences. Their agreement determined λ_p , from 0 for no retained detail to 1 for the full difference (Methods).

The model kept 78.1% of the neuron-specific difference on average in the demixed SST responses ($\lambda = 0.781$). It

Model construction and prediction

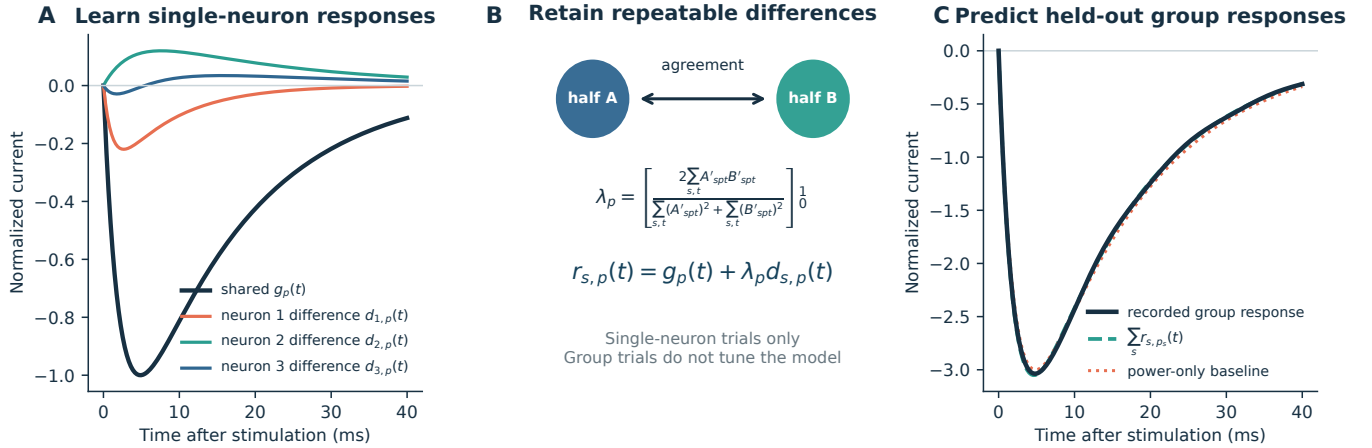


Figure 1: The prediction rule. **A**, At each laser power, the mean response to one neuron is split into a shared waveform $g_p(t)$ and a neuron-specific difference $d_{s,p}(t)$. Curves are schematic. **B**, Two alternating halves of the trials estimate reliability λ_p . Repeatable differences are kept; unstable differences are reduced. **C**, The model adds the learned responses of the stimulated neurons and tests the prediction on group trials that were not used for fitting.

improved both MSE (mean 8.92×10^{-4} , $q = 0.003906$) and waveform correlation (mean 0.00413, $q = 0.04688$) over the baseline. Both improvements remained positive when each file was removed in turn. Identity-shuffling controls passed, and replacing a stimulated neuron with a matched unstimulated neuron increased MSE in all nine files. These results helped develop the model, but they were not a held-out confirmation. They established the reliability-weighted rule to be tested in new data, while the PV files remained unopened under the original protocol.

2.4 The frozen rule generalized to a held-out E-to-E cohort

The prospective file-level test used all ten files in a checksum-verified E-to-E dataset. Before the data were opened, the protocol, file manifest, and SST development summary were locked with SHA-256 hashes. One file used the previously unused field name `caviar_weights_ensemble` instead of `caviar_weights_multi`. This naming difference was recorded before response values were read; it did not change the model.

The dataset contained 204,054 trials. The 129,613 single-neuron trials were used to learn the responses and reliability values. The 74,435 group trials were used only to test predictions, and every group trial was eligible. The reliability-weighted model improved demixed MSE over the baseline in all ten files (mean gain 2.1195×10^{-5} , exact $p = 0.0009766$, $q = 0.001953$) and waveform correlation in 9/10 files (mean gain 0.003973, $p = q = 0.008789$; Fig. 3, Table 4). The minimum improvement after re-

moving any one file was 5.62×10^{-6} for MSE and 0.00280 for waveform correlation. No single file determined either main result.

2.5 Biological grouping is not identified in the public files

The ten filenames identify recorded cells on three acquisition dates: five files from 14 December 2022, one from 16 December, and four from 19 December. The embedded field named `mouseID` contains labels such as `cell13_emx_map`; it does not identify animals or slices. The public files therefore support file-level comparisons across ten recorded cells but do not establish ten independent biological preparations.

As a conservative sensitivity analysis, file effects were averaged within the three dates before testing. The MSE and correlation effects remained positive on all three dates. With only three date groups, however, the smallest possible one-sided exact p value is 0.125. Acquisition date is not claimed to be an animal or slice identifier; this calculation shows how the inferential strength changes under a coarser plausible grouping. A verified file-to-slice-to-animal map is required for biological-level inference.

2.6 The prediction required the correct neuron identities

Improvement could arise because neuron-specific responses have a useful range of sizes, even when assigned to the wrong neurons. To test this possibility, neuron identities were shuffled within each imaging depth and laser power. The shuffle preserved the response set, group

Table 2: Dataset overview. Technical characteristics of the prospectively held-out E-to-E recordings used for the main file-level test.

Component	Recorded data	Role in the analysis
Preparation	Acute cortical slices; excitatory candidate neurons were stimulated while current was recorded from one patched excitatory neuron per file.	Tests excitatory-to-excitatory transfer of responses measured under controlled stimulation.
Recording units	10 files, each representing one labeled postsynaptic cell; 4,724 candidate-neuron entries, with 214–687 candidates per file.	File means are the inferential units because animal and slice identifiers are unavailable.
Spatial sampling	Candidates distributed across four or five two-photon imaging depths in each file; coordinates are stored in <code>targets</code> .	Depth is retained in identity-matched controls and null permutations.
Stimulation design	<code>stimulus_matrix</code> has one row per trial and one column per candidate neuron; positive entries give the delivered laser power.	Defines the stimulated set, group size, neuron identity, and exact source–power pairs.
Current traces	900 samples per trial at 20 kHz (45 ms), stored as raw current in <code>pscs</code> and NWD-demixed current in <code>pscs_demixed</code> .	Demixed current is the primary outcome; raw current provides a preprocessing sensitivity analysis.
Single-neuron trials	129,613 trials with exactly one positive stimulus entry.	Estimate shared and neuron-specific responses, reliability, source gain, and source latency.
Group trials	74,435 trials with more than one positive stimulus entry; all satisfy the minimum-data eligibility rule.	Held out from fitting and used only to evaluate predictions.
Blank trials	6 trials with no stimulated candidate and nonfinite current traces.	Excluded before response estimation or evaluation.
Biological metadata	10 recorded-cell labels across three acquisition dates; slice and animal membership are not provided.	Main tests are file-level; acquisition date is used only as a conservative clustering sensitivity.

sizes, powers, and recorded cell. Across 999 fixed repetitions per file, the observed MSE and correlation effects gave $z = 29.47$ and $z = 35.93$. Both were far above the familywise 99th-percentile threshold of $z = 2.93$ (both familywise $p = 0.001$; Fig. 4, Table 5).

A second control replaced each stimulated neuron with matched unstimulated neurons at the same depth and power. This replacement increased MSE in all ten files (mean cost 4.57×10^{-6} , exact $p = 0.0009766$). Together, these controls show that power, depth, response size, and group size were not enough. Improvement required assigning each response to the correct neuron.

2.7 Strength and latency explain the predictive improvement

The full model allows every neuron to have a different waveform. A simpler post-feedback model allowed only a neuron-specific scalar gain while forcing all neurons at one power to share the same temporal shape. The gain was fit by least-squares projection of each reliability-weighted single-neuron response onto the shared waveform. This model used no group responses.

Source-specific gain improved demixed MSE over the

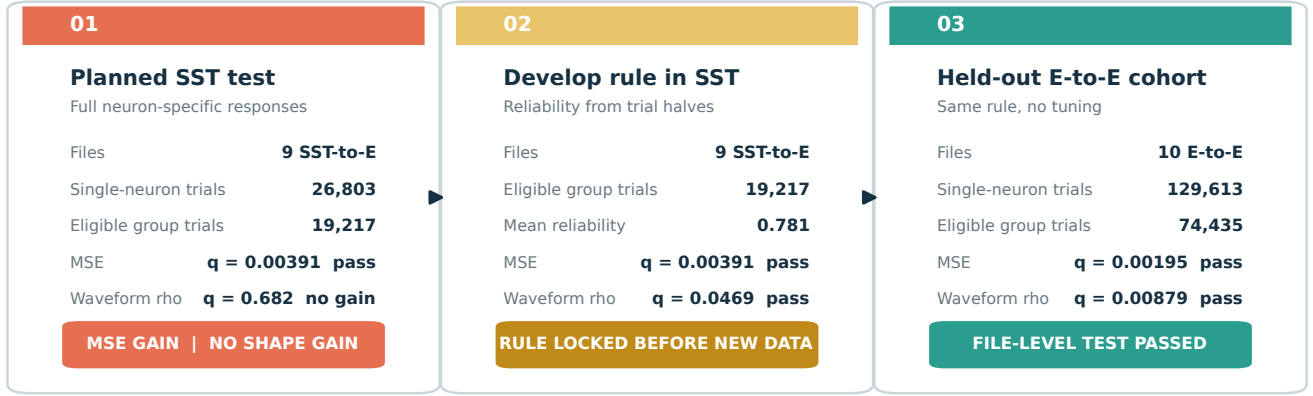
power-only baseline in all ten files (mean gain 2.051×10^{-5} , exact $p = 0.000977$). This is 96.7% of the full model’s mean MSE gain. Because a scalar does not change waveform correlation, the gain-only model had the same correlation as the power-only baseline. A second reduced model allowed the shared waveform to shift by up to 2 ms for each neuron. Gain plus latency recovered a mean correlation gain of 0.00190 and nearly all MSE improvement (2.119×10^{-5}).

Relative to gain plus latency, the full waveform changed mean MSE by only 6.27×10^{-9} ($p = 0.495$) and correlation by 0.00207 ($p = 0.104$). Thus the data clearly support source-specific response strength. They also suggest temporal information, but do not show that unrestricted waveform detail predicts groups better than the simpler gain-and-latency model (Fig. 5, Table 6).

2.8 Reliability changed with power and preparation

The model did not assume that every neuron-specific difference was reliable. Mean λ was 0.479 in the demixed E-to-E data, compared with 0.781 in the SST development data. In E-to-E files, mean reliability rose from

How the model was developed and tested



Datasets are not pooled for significance; the E-to-E result cannot revise the original SST/PV decision.

Figure 2: Order of the analysis. The first SST test improved MSE but not waveform correlation, so PV data remained unopened. The reliability was developed only in the already-open SST data. The rule and its success criteria were frozen before the E-to-E data were opened. Results are not pooled across stages or datasets.

Table 3: The three stages of the analysis. Effects are file-level means for the neuron-specific model relative to the baseline that knows laser power but not neuron identity. The reliability-weighted SST result was used for development, not held-out confirmation.

Model	Data	Role	MSE improvement	MSE q	Correlation improvement	Correlation q
Full response	SST	planned first test	8.777e-04	0.003906	-0.00263	0.6816
Reliability-weighted	SST	reliability development	8.922e-04	0.003906	0.00413	0.04688
Reliability-weighted	E-to-E	held-out cohort	2.120e-05	0.001953	0.00397	0.008789

0.264 at power 30 to 0.504 at power 45 and 0.670 at power 60 (Fig. 6). Two low-power estimates fell to zero, while one high-signal file kept more than 0.84 at every power. Predictions were therefore closer to the shared response when neuron-specific differences did not repeat, while repeatable differences were retained.

2.9 The reduced model reveals a raw-current latency effect

In the preregistered full-waveform comparison, two supporting measures, integrated-current error and peak-current error, improved in all ten E-to-E files (post-hoc three-outcome $q = 0.00146$ for each). Raw-current MSE also improved, but raw waveform correlation did not ($q = 0.803$). The claim about waveform shape therefore depends on NWD demixing.

Mean waveform correlation was 0.4109 for the neuron-specific model and 0.4070 for the baseline. Mean MSE was 6.1×10^{-5} for the neuron-specific model, 8.2×10^{-5} for the baseline, and 1.03×10^{-4} for zero. The file-level comparison with zero varied across preparations (4/10 positive, exact $p = 0.456$), as small currents make zero competitive under squared error.

The reduced-model sensitivity sharpened this result.

On raw current, source-specific gain improved MSE in all ten files but did not improve mean correlation. Gain plus latency improved both raw MSE and raw correlation in all ten files (mean correlation gain 0.00777, exact $p = 0.000977$). The unrestricted full waveform improved raw MSE further, but its mean correlation remained below the power-only baseline. Thus source-specific strength and latency both transfer to the direct physical recordings; arbitrary residual waveform detail does not.

3 Discussion

This reanalysis identifies a narrow, repeatable predictive effect. Responses learned from single-neuron stimulation improve predictions of held-out group current, and the improvement requires the correct neuron identities. The reduced-model comparison shows what carries most of that effect: some neurons produce reliably stronger post-synaptic responses than others at the same laser power. Those source-specific strengths remain useful when the neurons are stimulated in groups.

The temporal result is narrower than a general waveform claim. A source-specific latency shift improves correlation in both demixed and raw current, while the unre-

Prospectively held-out E-to-E cohort: results for each file

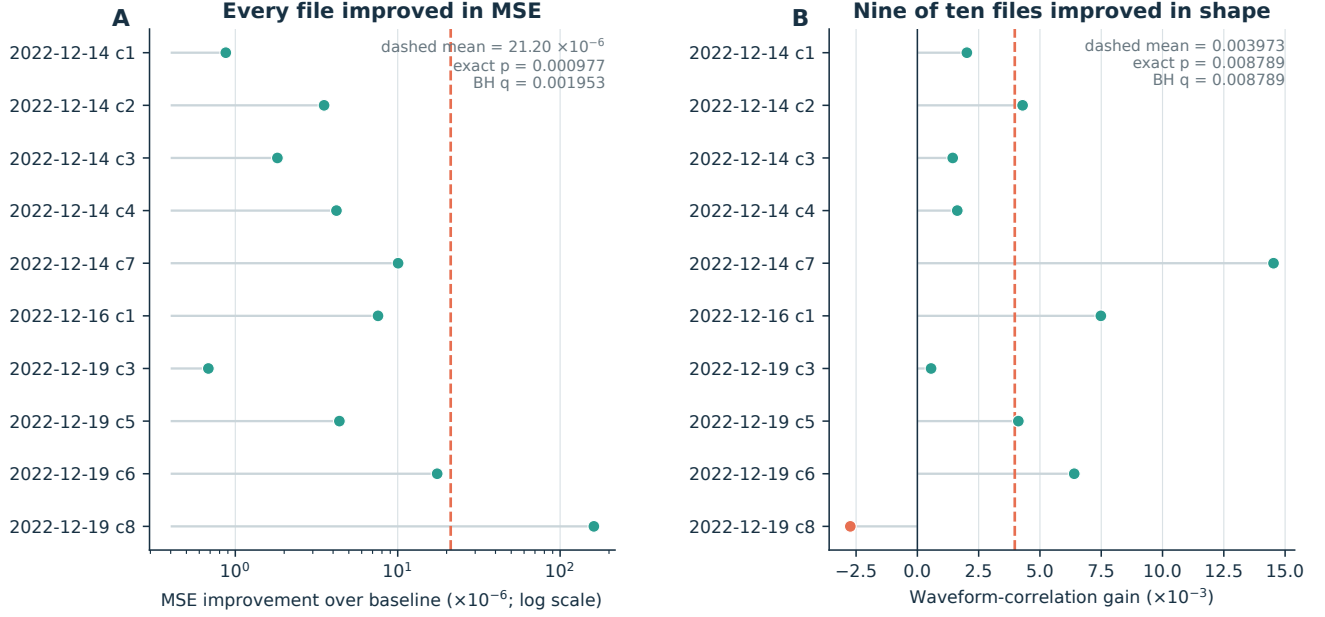


Figure 3: Main results in the new E-to-E data. **A**, MSE improves in all ten files when neuron identity is added to the baseline. The log axis shows the full range. **B**, Waveform correlation improves in nine files. Dashed lines show the mean across files. Statistical tests treat files, not individual trials, as the units of evidence. File independence at the animal or slice level cannot be established from the public metadata.

Table 4: Results of the prospectively held-out E-to-E test. LOO is the smallest mean effect after leaving out one file.

Measure	Mean improvement	Files improved	Exact p	Corrected q	Smallest LOO
MSE gain	2.120e-05	10/10	0.000977	0.001953	5.619e-06
Waveform $\Delta\rho$	0.00397	9/10	0.008789	0.008789	0.00280
MSE cost of using the wrong neuron	4.567e-06	10/10	0.000977	—	—

Table 5: Familywise tests of neuron identity.

Measure	Observed z	Familywise threshold	Familywise p
MSE gain	29.47	2.93	0.001
Waveform $\Delta\rho$	35.93	2.93	0.001

stricted waveform does not conclusively outperform gain plus latency. NWD does not receive neuron identity or fit the single-neuron templates used here, so the analysis is not a direct projection of group responses onto those templates. NWD is nevertheless trained on simulated PSC-like waveforms and favors clean evoked-current structure [Triplett et al., 2025]. Conclusions about residual waveform detail, beyond gain and latency, should therefore remain conditional on preprocessing.

The result differs from the leave-one-hologram-out prediction in the source study. CAVIaR learned from responses to other group holograms; the present models learn only from single-neuron trials. The new contribution is thus not that currents can be added or that an unseen hologram can be predicted. It is that source-specific

response strength and latency transfer from individual stimulation to withheld group stimulation without fitting any group response.

The prospective separation between SST development and E-to-E testing protects against adapting the rule to the E-to-E responses. It does not by itself establish biological independence. The ten files are ten recorded post-synaptic cells, but their animal and slice membership is absent from the public metadata. The three-date sensitivity preserves the direction of the effect but cannot provide a small exact p value. Biological-level inference requires the missing hierarchy.

3.1 Next steps and opportunities

The immediate next step is to obtain the anonymized mapping from files to slices and animals, then repeat the sign-flip tests and leave-one-out analysis at the highest independent biological level. This requires no new response analysis and directly determines the strength of the population-level claim.

Predictions require the correct neuron identity

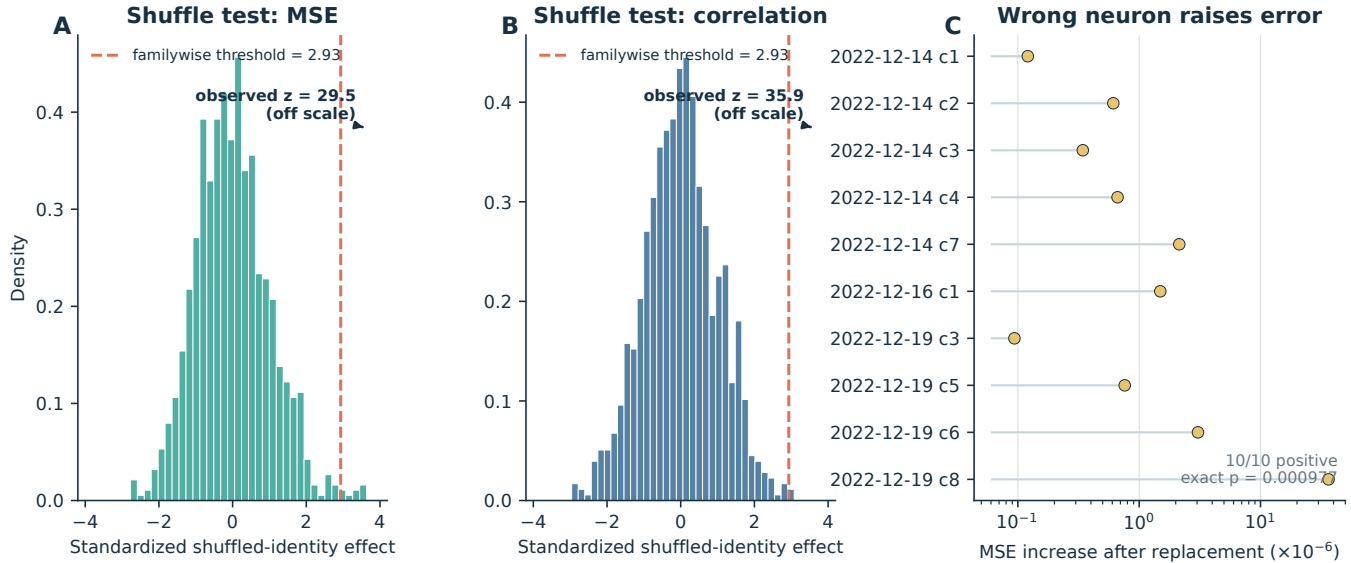


Figure 4: Controls for neuron identity. A–B, Null effects from 999 identity shuffles that preserve depth and power. The observed statistics are far beyond the plotted nulls and the familywise threshold. C, Replacing a stimulated neuron with matched unstimulated neurons increases MSE in every file.

Table 6: Post-feedback reduced-model comparison on demixed current. Gains are relative to the power-only model. All reduced-model parameters were learned from single-neuron trials only.

Single-neuron model	Mean MSE	MSE gain	Mean ρ	$\Delta\rho$
Power only	8.204e-05	—	0.4070	—
Source gain	6.153e-05	2.051e-05	0.4070	0.00000
Source gain + latency	6.085e-05	2.119e-05	0.4089	0.00190
Full waveform	6.084e-05	2.120e-05	0.4109	0.00397

A new preregistered comparison should then test the model hierarchy found here: power only, source gain, source gain plus latency, and unrestricted waveform. Applying that fixed hierarchy to another source-cell class would show whether latency or residual temporal detail generalizes beyond this post-feedback analysis.

The present models learn a separate response for every neuron and exact laser power. Holding out one neuron–power pair and predicting it from the same neuron at other powers would test whether source strength follows a rule across stimulation intensity rather than acting as an exact-power lookup.

A second experiment should be designed to distinguish addition from competing models. Repeated stimulation of selected pairs and progressively larger groups could compare the additive rule with saturation, pairwise interaction, and gain-control models on combinations withheld from fitting. The pattern of errors would then reveal not only whether additive predictions remain accurate, but

also where and how they give way to another response rule.

Temporal prediction should also be tested under recording conditions chosen for direct analysis of raw current. Longer intervals between stimuli, repeated presentations of the same groups, and source-blind filtering would reduce overlapping events and electrical noise without using neuron identity. These experiments would test whether residual waveform detail beyond strength and latency transfers to physical recordings.

Predictive models can also distinguish biological explanations. Direct measurements of presynaptic spiking could separate variable optical activation from variable synaptic transmission. Postsynaptic holding-potential changes and receptor-specific perturbations could test whether departures from addition arise from driving force, receptor recruitment, or recurrent interactions. Experiments should be selected so that candidate models make different predictions before the responses are observed.

Source-specific strength and latency explain the predictive gain

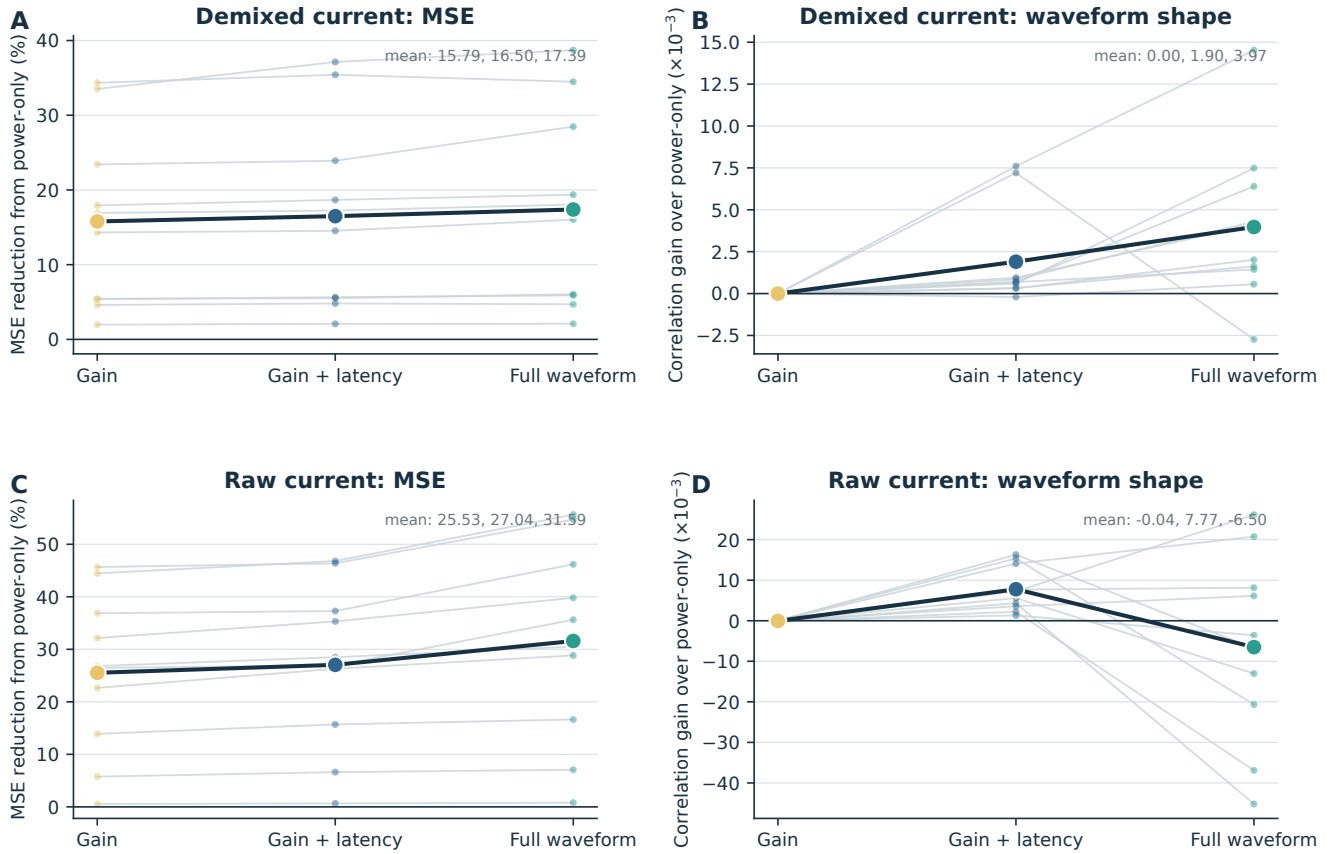


Figure 5: Which single-neuron information is needed? Thin lines show files and large points show equal-file means. MSE panels report percentage reduction from the power-only baseline; correlation panels report the absolute gain. **A–B**, NWD-demixed current. **C–D**, Raw current. Source-specific gain explains most MSE improvement. Gain plus latency improves correlation in demixed and raw current; unrestricted waveform detail does not add a conclusive advantage.

Together, the results support a compact predictive rule: source-specific response strength and latency learned from individual stimulation improve held-out group-current prediction. Unrestricted temporal detail remains an open test.

4 Methods

4.1 Public recordings and locked datasets

All recordings came from the CC BY 4.0 Figshare dataset “Neural circuit mapping using two-photon holographic optogenetics,” version 1 [Gajowa and Triplett, 2025], associated with Triplett et al. [2025]. The upstream analysis-code reference was frozen at `marcustriplett/circuitmap` commit `75934895f8ef02b8045ab1a8eee592a062c2489e`. The development manifest contained all nine SST-to-E files. The prospectively held-out manifest contained all ten E-to-E files. URLs, file sizes, and checksums

were recorded before analysis. Fourteen PV-to-E files remained unopened because the planned SST waveform criterion was not met.

Each MATLAB v7.3/HDF5 file contained a trial-by-neuron `stimulus_matrix`, trial-by-time raw and demixed current arrays (`pscs`, `pscs_demixed`), and neuron coordinates (`targets`). CAViAR weight fields were checked only to validate the file structure. They were not used to predict, filter, label, or select trials.

4.2 Learning trials, test trials, and response window

A single-neuron trial had exactly one positive stimulus entry. A group trial had more than one, and blank trials were excluded. Single-neuron trials were used only to learn responses and reliability. Group trials were used only for evaluation. Each trace had 900 samples at 20 kHz, with stimulation at sample 100. For each trial, the mean of samples 0–99 was subtracted and samples 120–899, or 1–40 ms after stimulation, were analyzed.

The data decide how much neuron-specific detail to keep

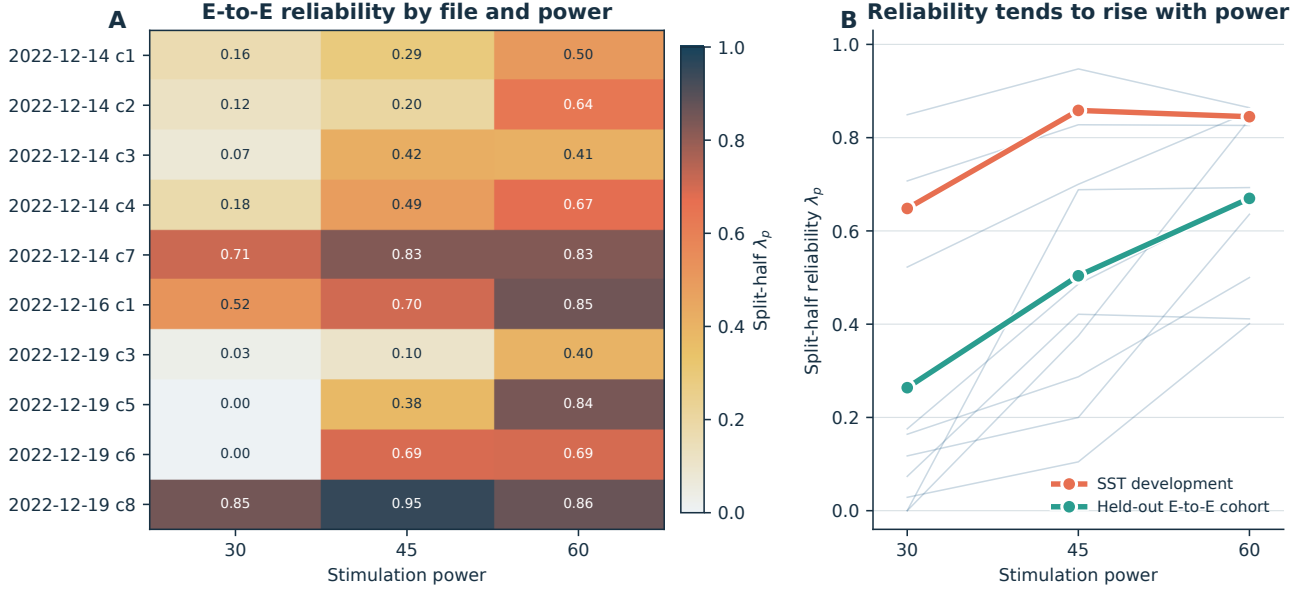


Figure 6: Reliability controls how much neuron-specific detail is kept. **A**, E-to-E reliability values by file and laser power. **B**, Thin lines show individual E-to-E files; thick lines show the mean across files. Reliability rises with power in E-to-E data and is lower than in the SST development data.

The first post-stimulus millisecond was excluded. NWD-demixed current was the main outcome; raw current was a supporting outcome.

A trace containing only nonfinite values was allowed only for a blank trial, which was then excluded. Any partly nonfinite trace or nonfinite stimulated trial failed validation. Laser powers were not grouped into bins. Learning a response for one neuron at one power required at least three single-neuron trials. A group trial was eligible only when every stimulated neuron met this requirement at its exact delivered power.

4.3 Neuron-specific and baseline models

Let $\bar{K}_{s,p}(t)$ be the mean response across single-neuron trials for neuron s at exact power p . The baseline response was the mean across neurons, giving every neuron equal weight rather than weighting by its number of trials:

$$g_p(t) = \frac{1}{|S_p|} \sum_{s \in S_p} \bar{K}_{s,p}(t), \quad (3)$$

The neuron-specific difference was $d_{s,p}(t) = \bar{K}_{s,p}(t) - g_p(t)$.

To estimate reliability, trials for each (s,p) pair were assigned in stored order to alternating halves. Let $A_{s,p}(t)$ and $B_{s,p}(t)$ be the two half means. Each half was centered independently across neurons: $A'_{s,p} = A_{s,p} - |S_p|^{-1} \sum_s A_{s,p}$ and analogously for B' . For each file, out-

come, and power,

$$\lambda_p = \left[\frac{2 \sum_{s,t} A'_{s,p}(t) B'_{s,p}(t)}{\sum_{s,t} A'^2_{s,p}(t) + \sum_{s,t} B'^2_{s,p}(t)} \right]_0^1. \quad (4)$$

The brackets limit the value to $[0, 1]$. After determining λ_p , the final response followed Eq. 1 and used the mean of all single-neuron trials.

For an eligible group E , the baseline added $g_{p_s}(t)$ for the stimulated neurons. The neuron-specific model used Eq. 2. A zero waveform was included as a supporting baseline. Group responses were not used to fit an intercept, coefficient, gain, latency shift, saturation term, time window, rank, or neuron filter.

4.4 Reduced models added after external feedback

The gain-only model was constructed from the frozen reliability-weighted response $r_{s,p}$ and the shared waveform g_p . For every source and power, the least-squares scalar was

$$\alpha_{s,p} = \left[\frac{\langle r_{s,p}, g_p \rangle}{\langle g_p, g_p \rangle} \right]_0^\infty, \quad (5)$$

and the reduced response was $\alpha_{s,p} g_p(t)$. Negative projections were set to zero; if g_p had zero energy, $\alpha_{s,p}$ was set to one. The gain-plus-latency model shifted g_p by each integer lag from -40 to 40 samples (± 2 ms), refit α for each lag, and retained the pair with the smallest single-neuron waveform error. No group response selected a gain, lag, or model.

These reduced models and their comparisons were specified after external feedback and are reported as sensitivity analyses, not as part of the locked E-to-E gate. Exact sign-flip tests used the same equal-file aggregation as the original analysis.

4.5 Biological-unit audit and date sensitivity

The filename supplied the acquisition date and recorded-cell label. The MATLAB `ExpStruct.mouseID` field was read directly from every E-to-E file, but its values repeated cell labels rather than animal identifiers. No slice identifier was found. To show the consequence of possible clustering, file effects were averaged within acquisition date and the exact sign-flip test was repeated across the three dates. Date was used only as a conservative session proxy and was not interpreted as a verified animal or slice.

4.6 Measurements and statistical tests

Four measures were calculated for each eligible trial: response-window MSE, Pearson waveform correlation, absolute error in integrated current, and absolute error in peak current. Correlation was set to zero when either waveform was constant. Integrated current was the response-window sum multiplied by 0.05 ms. Peak current was the signed sample with the largest absolute value.

The two main effects were baseline MSE minus neuron-specific MSE and neuron-specific correlation minus baseline correlation. Trial-level effects were first averaged within each file. Across files, exact one-sided sign-flip tests considered all 2^n sign assignments [Fisher, 1935]. The two main p values were corrected by Benjamini–Hochberg [Benjamini and Hochberg, 1995]. Both equal-file means had to be positive with $q < 0.05$. Each effect also had to remain positive after leaving out any one file.

4.7 Identity shuffling and matched-neuron replacement

For each of 999 fixed null repetitions per file, neuron identities were shuffled separately within imaging depth and exact power. The same shuffle was applied to all group trials in one repetition. This kept power, depth, group size, recorded cell, and the set of learned responses unchanged. Null effects gave every file equal weight. Because MSE and correlation use different units, each was standardized using its own null mean and standard deviation. The larger standardized value from the two measures formed a familywise null. Both observed z scores had to exceed its 99th percentile.

The replacement test asked whether each correct identity was necessary. Each stimulated neuron was replaced in turn by every eligible unstimulated neuron at the same

depth and exact power. Replacement MSE minus the original MSE was averaged across replacements, stimulated neurons, trials, and then files. A positive exact one-sided sign-flip test across files was required.

4.8 Reproducibility

The public repository contains frozen protocols and lock records, raw-data manifests, analysis and download code, unit tests, compact derived tables, figure builders, and this manuscript. All figures are generated from the included CSV and JSON outputs. Reproducing the complete analysis requires downloading about 2.01 GiB of SST recordings and 7.64 GiB of E-to-E recordings. The repository does not redistribute the raw recordings.

Data and code availability

Source data are available from Figshare at <https://doi.org/10.6084/m9.figshare.25641435.v1>. The reproducible manuscript repository is <https://github.com/asim-v/holographic-source-grammar>. Code is released under MIT; original manuscript materials and derived tables are CC BY 4.0.

Ethics statement

This study is a secondary analysis of public data and introduced no new animal experiments. Animal procedures and approvals are described in the source publication [Triplett et al., 2025].

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Author contributions

J.E.B.S. conceived the reanalysis, designed the protocols, implemented and validated the software, analyzed the data, prepared the visualizations, and wrote the manuscript.

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Competing interests

The author declares no competing interests.

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